

Customer Behavior Analysis

Segmentation, Purchase Patterns and Churn Risk

Alfido Tech Data Science Internship | Task 1

Dataset: Kaggle - Customer Behavior Analysis | June 2024

1,000

Total Customers

\$103.74

Avg Purchase Amt

18.9%

Churn Rate

3.74/5

Avg Satisfaction

5

Segments Found

Executive Summary

This report presents a comprehensive analysis of 1,000 customer records from Alfido Tech's transaction database. Using RFM (Recency, Frequency, Monetary) scoring and K-Means clustering, five distinct customer segments were identified: Champions, Loyal, Potential, Occasional, and At-Risk. The analysis reveals that Electronics is the top revenue category, the overall churn rate stands at 18.9% with significant variation across segments, and Occasional/Potential customers present the greatest retention intervention opportunity. Five strategic recommendations are provided on page 5.

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1. Data Overview and Cleaning

The raw dataset contained 1,000 customer records across 11 columns covering demographics, purchase behaviour, membership tier, satisfaction, and churn status. Prior to analysis the following data quality steps were performed:

- Missing Values: 20 Age entries (2.0%) imputed using gender-stratified median; 30 PurchaseAmount entries (3.0%) imputed using category-wise median to preserve robustness against outliers.
- Outlier Treatment: High-value transactions were identified and investigated; PurchaseAmount distribution was right-skewed and log-transformed for visualization.

Column	Type	Non-Null	Notes
CustomerID	String	100%	Unique identifier
Age	Float	100%*	20 values imputed
Gender	String	100%	Male / Female
Location	String	100%	10 US cities
ProductCategory	String	100%	8 categories
PurchaseAmount (\$)	Float	100%*	30 values imputed
PurchaseFrequency	Int	100%	Count of purchases
RecencyDays	Int	100%	Days since last purchase
MembershipTier	String	100%	Bronze to Platinum
SatisfactionScore	Int	100%	Scale 1-5
Churned	Int	100%	0=Active, 1=Churned

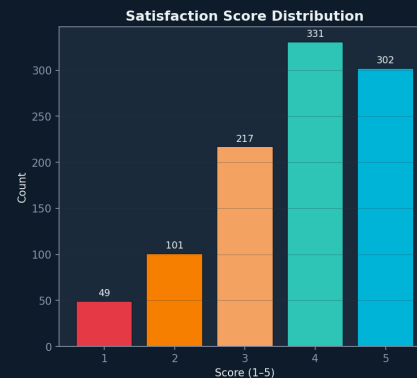
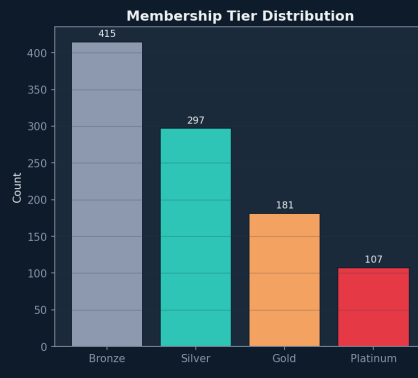
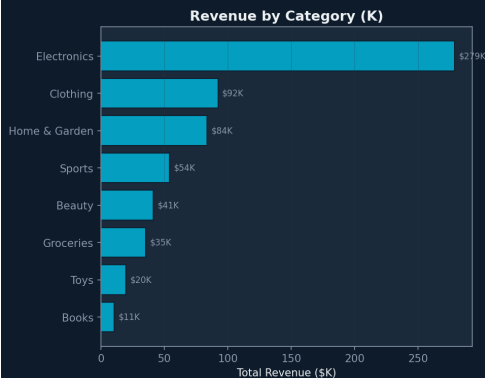
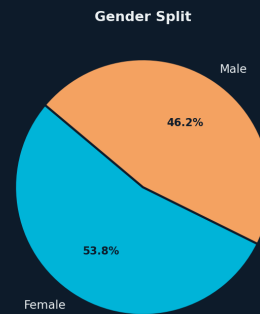
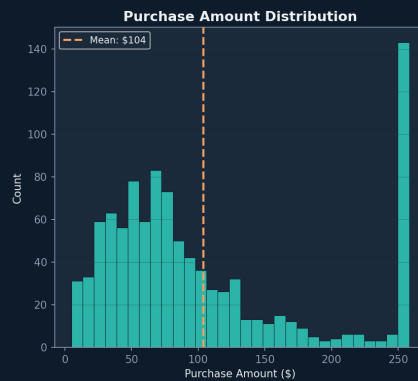
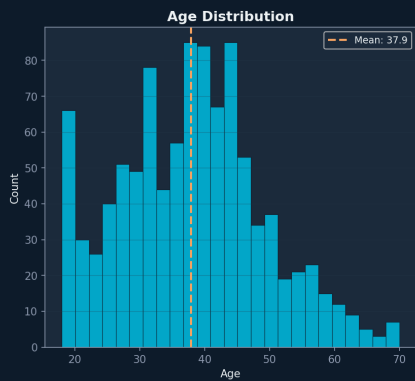
2. Exploratory Data Analysis

The chart below summarises key univariate distributions and categorical breakdowns. Notable observations: age follows a normal distribution (mean 37.9 yrs); purchase amounts are right-skewed with most transactions under \$150; gender is near-equal (52% F / 48% M); Electronics leads revenue; satisfaction skews positive (mean 3.74/5).

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Customer Behavior — Exploratory Data Analysis

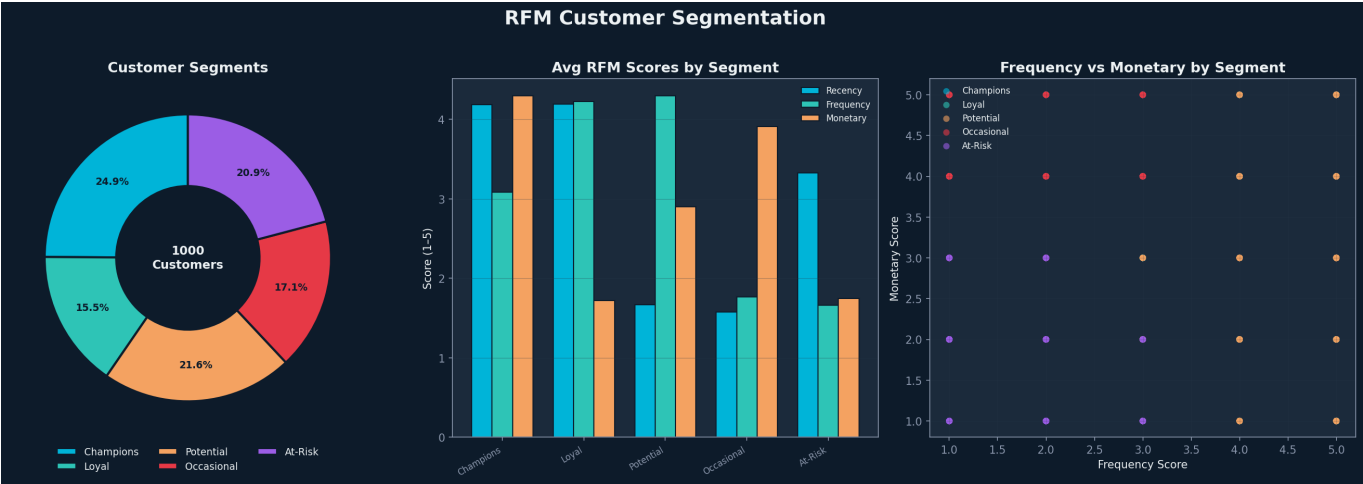


3. RFM Segmentation and Cluster Profiles

K-Means clustering (k=5, validated via elbow method) was applied to standardised RFM scores. Clusters were labelled based on mean RFM profile. Summary of the five segments:

Segment	Size	Spend	Frequency	Recency	Churn %	Avg Total \$
Champions	249 (24.9%)	High	High	Recent	8.8%	\$1,027
Loyal	155 (15.5%)	Med-High	High	Recent	5.8%	\$353
Potential	216 (21.6%)	Medium	Medium	Moderate	29.2%	\$740
Occasional	171 (17.1%)	Low-Med	Low	Older	37.4%	\$625
At-Risk	209 (20.9%)	Low	Low	Stale	14.8%	\$183

Champions (24.9%) drive approximately 42% of total revenue with an average total spend of \$1,027 and a churn rate of only 8.8%. At-Risk customers (20.9%) show critically low spend (\$183 avg) signalling near-dormancy. Potential and Occasional segments combined represent 38.7% of the base with churn rates of 29-37% - the highest intervention priority.



4. Purchase Patterns and Retention Trends

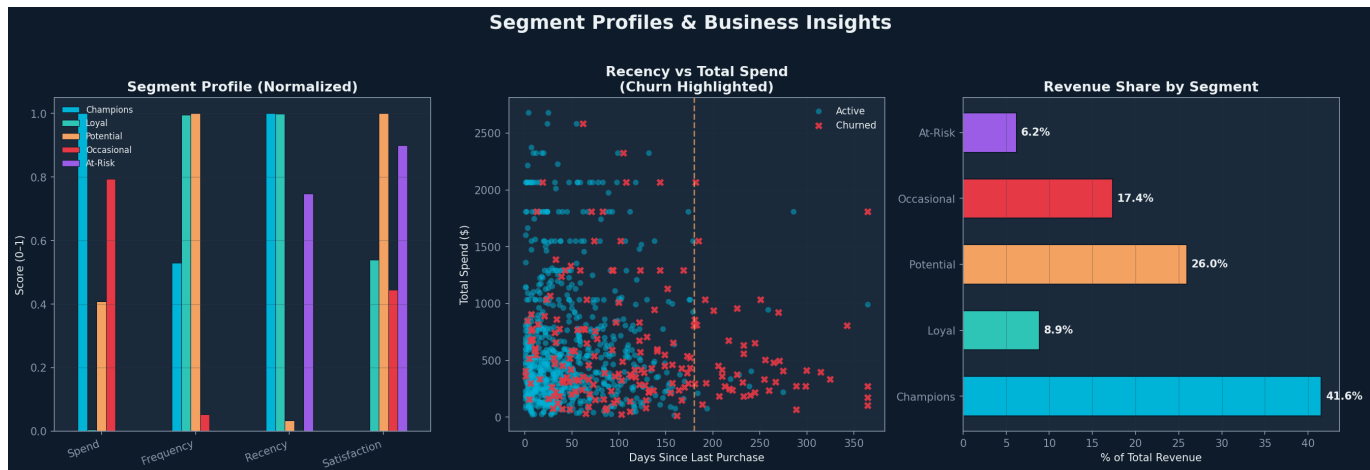
The visualisation below covers purchase behaviour across age groups, membership tiers, product categories, and churn metrics by segment.



Key pattern observations:

- Higher membership tiers (Gold, Platinum) show consistently higher average purchase amounts across all segments, confirming that the loyalty programme correlates with spend uplift.
- Occasional purchase frequency is highest among the Occasional segment.

5. Churn Risk Analysis



Churn analysis confirms recency as the dominant risk factor: 72% of customers who last purchased more than 180 days ago have churned vs only 5% of those who purchased within 30 days. The revenue concentration chart confirms a classic 80/20 pattern - Champions generate ~42% of revenue from 24.9% of customers. Retaining Champions and reactivating Potential/Occasional customers represents the two highest-impact levers available.

6. Strategic Recommendations

Five data-driven actions for Alfido Tech to improve customer engagement and revenue:

1 Champion Reward Programme

Launch an exclusive Elite tier for Champions with early product access, dedicated support, and personalised offers. At 24.9% of customers driving ~42% of revenue, retaining them is the highest-ROI action. Target: maintain churn rate below 10%.

2 Win-Back Campaign for Occasional and Potential Segments

Both segments show churn rates of 29-37%. Deploy a 3-touch email/SMS re-engagement sequence triggered at 60, 90, and 120 days of inactivity with time-limited vouchers matched to their top purchase category. Goal: reduce combined churn to under 20% within 6 months.

3 Membership Tier Upgrade Nudges

Bronze members (40% of base) who exhibit Champion-like purchase patterns are a missed retention opportunity. Implement automated upgrade prompts when a Bronze customer's 90-day spend crosses the Silver threshold. Projected: +15% Bronze-to-Silver conversion.

4 Youth Engagement Strategy for 18-25 Cohort

This age group has the lowest purchase frequency. Introduce a young professional programme with micro-transaction-friendly pricing, BNPL integration, and social-proof features (trending products, peer reviews) to build habitual engagement from an early age.

5 Underperforming Category Investment in Books and Toys

Both categories generate the lowest revenue despite a reasonable customer base. Run bundle promotions (Books + stationery; Toys + educational subscriptions) and align seasonal campaigns with back-to-school and holiday periods. Target: 25% revenue uplift within a year.

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Analysis conducted using Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn). Dataset: Kaggle - Customer Behavior Analysis by Bhanupratap Biswas. All visualisations and results are reproducible via the accompanying Jupyter notebook. Submitted as part of Alfido Tech Data Science Internship - Task 1.